Stein's core equations

Setup. $X \sim N(\theta, \sigma^2 I_p)$, with θ unknown. Any estimator can be written as $\hat{\theta} = X + g(X)$.

1. Stein's lemma

$$\mathbb{E}\big[(X_i - \theta_i)\, g_i(X)\big] \;=\; \sigma^2\, \mathbb{E}\left[\frac{\partial g_i(X)}{\partial X_i}\right]$$

The left side involves the unknown θ. The right side needs only a derivative of g, which is computable from the data.

2. SURE (Stein's Unbiased Risk Estimate)

$$\mathbb{E}\,\|\hat{\theta} - \theta\|^2 \;=\; \mathbb{E}\Big[\; \underbrace{p\sigma^2}_{\text{risk of } X} \;+\; \underbrace{\|g(X)\|^2}_{\text{how far you moved}} \;+\; \underbrace{2\sigma^2 \sum_i \frac{\partial g_i(X)}{\partial X_i}}_{\text{cross term, via Stein's lemma}} \;\Big]$$

Everything inside the brackets is computable from X, so it is an unbiased estimate of the risk.

3. The version that carries over to the thesis

Before applying Stein's lemma, the same identity reads

$$\mathbb{E}\,\|\hat{\theta} - \theta\|^2 \;=\; \mathbb{E}\,\|\hat{\theta} - X\|^2 \;-\; p\sigma^2 \;+\; 2\sum_i \mathrm{Cov}(\hat{\theta}_i, X_i)$$

Substitute $X \to S$ and $\theta \to \Sigma$:

$$\mathbb{E}\,\|\hat{\Sigma} - \Sigma\|_F^2 \;=\; \mathbb{E}\,\|\hat{\Sigma} - S\|_F^2 \;-\; \sum_{i,j} \mathrm{Var}(s_{ij}) \;+\; 2\sum_{i,j} \mathrm{Cov}(\hat{\sigma}_{ij}, s_{ij})$$

This is Yi & Zou's Lemma 1. Stein's lemma is what turns the Cov term into a derivative, and that step requires Gaussian data.

Ledoit–Wolf shrinkage

The problem. You want a $p \times p$ covariance matrix Σ from T observations. The sample covariance S is unbiased, but when p is comparable to T it is very noisy: the large eigenvalues come out too large, the small ones too small, and the inverse is unstable. A structured target F (for example the identity times a scalar) has little noise but is biased. LW combine the two.

The estimator.

$$\hat{\Sigma} \;=\; \delta F + (1-\delta)S, \qquad 0 \le \delta \le 1$$

The *shrinkage intensity* δ is the knob. It is the same idea as James–Stein: pull a noisy unbiased estimate toward a simple target, and accept a little bias for a large cut in variance.

The loss. Frobenius risk, i.e. total squared error over all p^2 entries:

$$R(\delta) = \mathbb{E}\,\|\hat{\Sigma} - \Sigma\|_F^2 = \sum_{i,j} \mathbb{E}\big(\hat{\sigma}_{ij} - \sigma_{ij}\big)^2$$

1. The optimal intensity (exact, finite sample)

Each entry's error is variance plus squared bias. Using $\mathbb{E}s_{ij} = \sigma_{ij}$ and $\phi_{ij} = \mathbb{E}f_{ij}$, minimizing $R(\delta)$ gives

$$\delta^* \;=\; \frac{\displaystyle\sum_{i,j}\Big[\operatorname{Var}(s_{ij}) - \operatorname{Cov}(f_{ij}, s_{ij})\Big]}{\displaystyle\sum_{i,j}\Big[\operatorname{Var}(f_{ij} - s_{ij}) + (\phi_{ij} - \sigma_{ij})^2\Big]} \;=\; \frac{\text{noise in } S \;-\; \text{co-movement of } F \text{ with } S}{\mathbb{E}\,\|F - S\|_F^2}$$

- **Numerator:** how noisy S is. More noise means shrink more. The Cov term subtracts the part of the noise that F shares with S, because F is built from the same data and cannot cancel that part.
- **Denominator:** how far apart F and S are, which includes the bias of the target $(\phi_{ij} - \sigma_{ij})^2$. A badly wrong target means shrink less.

This δ^* is an *oracle*: it depends on the unknown Σ.

2. LW's move: asymptotic approximation

LW write $\delta^* \approx \kappa/T$ with

$$\kappa = \frac{\pi - \rho}{\gamma}, \qquad \pi = \sum_{i,j} \operatorname{AsyVar}(\sqrt{T}\, s_{ij}), \quad \rho = \sum_{i,j} \operatorname{AsyCov}(\sqrt{T}\, f_{ij}, \sqrt{T}\, s_{ij}), \quad \gamma = \sum_{i,j} (\phi_{ij} - \sigma_{ij})^2$$

and plug in *consistent* estimators $\hat{\pi}, \hat{\rho}, \hat{\gamma}$ (with $\hat{\gamma}$ built from $\|F - S\|_F^2$):

$$\hat{\delta} = \max\Big\{0,\ \min\Big\{\frac{\hat{\kappa}}{T},\ 1\Big\}\Big\}$$

Targets. LW (2003): single-index (market model) $F = s_m^2\,\hat{\beta}\hat{\beta}^\top + D$. LW (2004): scaled identity $F = \nu I$ with $\nu = \operatorname{tr} S/p$, which gives the closed form

$$\hat{\delta}_{\mathrm{LW}} = \frac{\frac{1}{T^2}\sum_t \|x_t x_t^\top - S\|_F^2}{\|S - \nu I\|_F^2}$$

2004 intuition (a Pythagorean identity). With $\mu = \operatorname{tr}\Sigma/p$,

$$\underbrace{\mathbb{E}\|S - \mu I\|^2}_{d^2\text{: spread of } S} = \underbrace{\|\Sigma - \mu I\|^2}_{a^2\text{: true spread}} + \underbrace{\mathbb{E}\|S - \Sigma\|^2}_{b^2\text{: estimation error}}, \qquad \delta^* = \frac{b^2}{d^2}$$

The share of S's apparent dispersion that is pure noise is exactly how much to shrink.

3. Link to Stein and to the thesis

Plug $\hat{\Sigma} = \delta F + (1-\delta)S$ into the covariance form of SURE (page 1, eq. 3):

$$R(\delta) = \delta^2\,\mathbb{E}\|F - S\|_F^2 + (1 - 2\delta)\sum_{i,j}\operatorname{Var}(s_{ij}) + 2\delta\sum_{i,j}\operatorname{Cov}(f_{ij}, s_{ij})$$

Its minimizer is exactly LW's δ^* above. So LW and SURE describe the *same* risk curve.

Ledoit–Wolf	**SURE (thesis)**
estimate the three terms *consistently* ($T \to \infty$)	estimate them *unbiasedly* (any T)
gives one number $\hat{\delta}$	gives the whole curve $\widehat{R}(\delta)$
Cov term via asymptotics	Cov term exact when F is linear in S

The thesis question: does finite-sample unbiasedness beat asymptotic consistency when T is small relative to p, and what happens for targets (single-index) where the Cov term has no unbiased estimate?

Two Stein shrinkers, worked through SURE

Same setup as page 1: $X \sim N(\theta, \sigma^2 I_p)$, $\hat\theta = X + g(X)$.

1. Linear shrinkage, $\hat\theta = (1-\lambda)X$

Here $g = -\lambda X$ and $\sum_i \partial g_i/\partial X_i = -\lambda p$, so

$$\mathrm{SURE}(\lambda) = p\sigma^2 + \lambda^2\|X\|^2 - 2\lambda p\sigma^2, \qquad \hat\lambda = \frac{p\sigma^2}{\|X\|^2}.$$

The true risk is $R(\lambda) = (1-\lambda)^2 p\sigma^2 + \lambda^2\|\theta\|^2$, minimized at $\lambda^* = p\sigma^2/(p\sigma^2 + \|\theta\|^2)$. Since $\mathbb{E}\|X\|^2 = p\sigma^2 + \|\theta\|^2$, $\hat\lambda$ is a plug-in ratio for λ^*, the same shape as $\hat\delta$ vs. δ^* on page 2.

2. James–Stein, $\hat\theta = \big(1 - c/\|X\|^2\big)X$

Here $\sum_i \partial g_i/\partial X_i = -c(p-2)/\|X\|^2$, so

$$\mathrm{SURE}(c) = p\sigma^2 + \frac{c^2 - 2c(p-2)\sigma^2}{\|X\|^2},$$

minimized at $c = (p-2)\sigma^2$, giving $\mathrm{SURE} = p\sigma^2 - (p-2)^2\sigma^4/\|X\|^2 < p\sigma^2$ for $p \geq 3$. So James–Stein beats X for every θ. At $\theta = 0$ (with $\sigma = 1$) its risk is $p - (p-2)^2\,\mathbb{E}[1/\chi^2_p] = 2$.

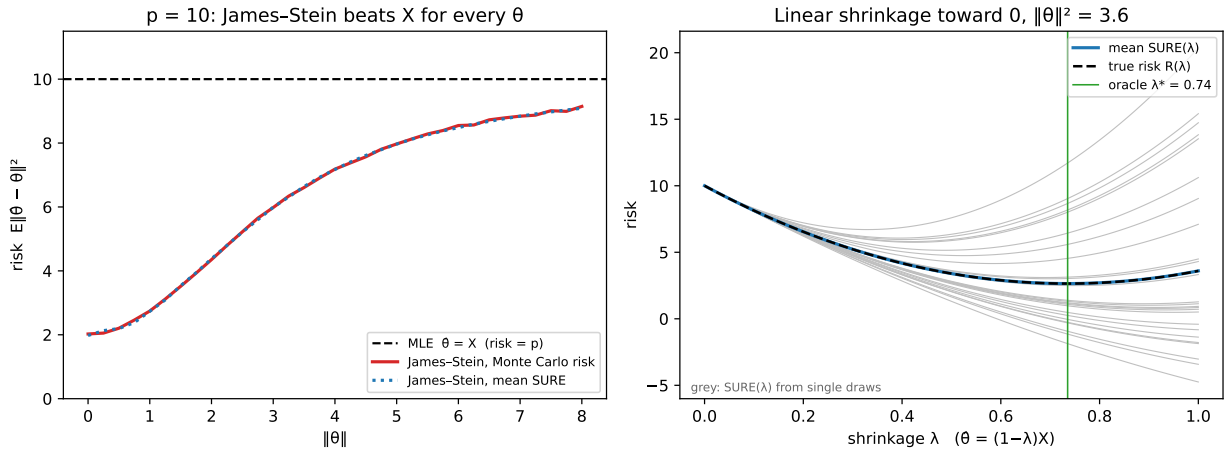


Figure (`sims/stein_js.py`, $p = 10$, $\sigma = 1$). Left: risk of the MLE and of James–Stein as $\|\theta\|$ grows, with the mean SURE of James–Stein. Right: $\mathrm{SURE}(\lambda)$ for linear shrinkage from single draws (grey), its mean, and the true risk.

Deriving the three-piece formula, step by step

Goal. Show that for *any* estimate $\hat{\Sigma}$, if $\mathbb{E}S = \Sigma$,

$$\mathbb{E}\|\hat{\Sigma} - \Sigma\|_F^2 = \underbrace{\mathbb{E}\|\hat{\Sigma} - S\|_F^2}_{\text{(a)}} - \underbrace{V}_{\text{(b)}} + \underbrace{2\sum_{i,j} \mathrm{Cov}(\hat{\sigma}_{ij}, s_{ij})}_{\text{(c)}}, \qquad V = \sum_{i,j} \mathrm{Var}(s_{ij}).$$

Step 1. Turn the matrix norm into a sum of numbers

The squared Frobenius norm is the sum of squared entries, $\|A\|_F^2 = \sum_{i,j} a_{ij}^2$. So

$$\mathbb{E}\|\hat{\Sigma} - \Sigma\|_F^2 = \sum_{i,j} \mathbb{E}\big(\hat{\sigma}_{ij} - \sigma_{ij}\big)^2.$$

From here on, fix one entry (i, j) and drop the subscripts: write $\hat{\sigma},\ s,\ \sigma$. We only need to handle one number at a time, then add up.

Step 2. Add and subtract s

$$\hat{\sigma} - \sigma = \underbrace{(\hat{\sigma} - s)}_{\text{we can see this}} + \underbrace{(s - \sigma)}_{\text{noise in } s}$$

This is the whole trick: split the unknown error into a part we observe and the noise of the data.

Step 3. Square it

Using $(u + w)^2 = u^2 + w^2 + 2uw$:

$$(\hat{\sigma} - \sigma)^2 = (\hat{\sigma} - s)^2 + (s - \sigma)^2 + 2(\hat{\sigma} - s)(s - \sigma).$$

Step 4. Take expectations, term by term

- First term: $\mathbb{E}(\hat{\sigma} - s)^2$. Keep it as is. Summed over (i, j) this is (a).
- Second term: $\mathbb{E}(s - \sigma)^2 = \mathrm{Var}(s)$, *because* $\mathbb{E}s = \sigma$ (variance is the expected squared distance from the mean). This is the first place we use $\mathbb{E}S = \Sigma$.
- Third term: the cross term $2\,\mathbb{E}\big[(\hat{\sigma} - s)(s - \sigma)\big]$, handled in Step 5.

Step 5. Simplify the cross term

Split it in two:

$$\mathbb{E}\big[(\hat{\sigma} - s)(s - \sigma)\big] = \mathbb{E}\big[\hat{\sigma}\,(s - \sigma)\big] - \mathbb{E}\big[s\,(s - \sigma)\big].$$

Why $\mathbb{E}[\hat{\sigma}(s-\sigma)] = \mathrm{Cov}(\hat{\sigma}, s)$. Start from the definition of covariance for any two random numbers A and B:

$$\mathrm{Cov}(A, B) = \mathbb{E}\big[(A - \mathbb{E}A)(B - \mathbb{E}B)\big].$$

Multiply out the first bracket:

$$\mathrm{Cov}(A, B) = \mathbb{E}\big[A\,(B - \mathbb{E}B)\big] - \mathbb{E}A \cdot \underbrace{\mathbb{E}\big[B - \mathbb{E}B\big]}_{=\mathbb{E}B - \mathbb{E}B\,=\,0} = \mathbb{E}\big[A\,(B - \mathbb{E}B)\big].$$

($\mathbb{E}A$ is a constant, so it comes out of the expectation.) In words: to get a covariance you only need to centre *one* of the two variables; centring the other adds a term with mean zero. Now take $A = \hat{\sigma}$ and $B = s$. Since $\mathbb{E}s = \sigma$, the centred B is exactly $s - \sigma$:

$$\mathrm{Cov}(\hat{\sigma}, s) = \mathbb{E}\big[\hat{\sigma}\,(s - \mathbb{E}s)\big] = \mathbb{E}\big[\hat{\sigma}\,(s - \sigma)\big].$$

This is where $\mathbb{E}S = \Sigma$ is needed: if $\mathbb{E}s \neq \sigma$, then $s - \sigma$ is not centred and the identity picks up an extra $\mathbb{E}\hat{\sigma} \cdot (\mathbb{E}s - \sigma)$. Note that $\hat{\sigma}$ itself never needs to be centred, and $\hat{\sigma}$ can be biased, data-dependent, anything.

Why $\mathbb{E}[s(s-\sigma)] = \mathrm{Var}(s)$. Same identity with $A = s$: $\mathbb{E}[s(s-\sigma)] = \mathrm{Cov}(s, s) = \mathrm{Var}(s)$. So the cross term is $2\,\mathrm{Cov}(\hat{\sigma}, s) - 2\,\mathrm{Var}(s)$. This is the second place we use $\mathbb{E}S = \Sigma$.

Step 6. Put the pieces back together

$$\mathbb{E}(\hat{\sigma} - \sigma)^2 = \mathbb{E}(\hat{\sigma} - s)^2 + \mathrm{Var}(s) + 2\,\mathrm{Cov}(\hat{\sigma}, s) - 2\,\mathrm{Var}(s) = \mathbb{E}(\hat{\sigma} - s)^2 - \mathrm{Var}(s) + 2\,\mathrm{Cov}(\hat{\sigma}, s).$$

Summing over all (i, j) gives exactly (a) − (b) + (c). □

What the derivation used, and what it did not

- **Used:** only $\mathbb{E}S = \Sigma$, in Steps 4 and 5. This is why the open issue about dividing by T vs. $T - 1$ (and the unknown mean) matters: if $\mathbb{E}S \neq \Sigma$, a bias term appears.
- **Not used:** any distribution (no normality), and nothing about the form of $\hat{\Sigma}$. It holds for LW, for tapering (Yi & Zou), for any estimator.
- **Stein's version** is the same six steps with $X, \theta, \hat{\theta}$ in place of $S, \Sigma, \hat{\Sigma}$. Stein then adds one more step that needs normality: $\mathrm{Cov}(\hat{\theta}_i, X_i) = \sigma^2\,\mathbb{E}[\partial\hat{\theta}_i / \partial X_i]$. The thesis replaces that step with U-statistics (pages 8–9).

The covariance version, in plain words

The idea in one line. We want to know how wrong our covariance estimate $\hat{\Sigma}$ is, but the truth Σ is unknown. Stein's trick, carried over to matrices, rewrites "how wrong" using only things the data can tell us.

1. Split the error into three pieces

For any estimate $\hat{\Sigma}$, as long as S is right on average ($\mathbb{E}S = \Sigma$):

$$\underbrace{\mathbb{E}\|\hat{\Sigma} - \Sigma\|_F^2}_{\text{how wrong we are}} = \underbrace{\mathbb{E}\|\hat{\Sigma} - S\|_F^2}_{\text{(a) how far we moved from } S} - \underbrace{V}_{\text{(b) noise in } S} + \underbrace{2\sum_{i,j} \mathrm{Cov}(\hat{\sigma}_{ij}, s_{ij})}_{\text{(c) how much we copy that noise}}$$

(a) Observable: we can compute $\|\hat{\Sigma} - S\|_F^2$ directly.

(b) $V = \sum_{ij} \mathrm{Var}(s_{ij}) = \mathbb{E}\|S - \Sigma\|_F^2$, the error of the raw sample covariance. It is subtracted because part of (a) is just S being noisy, not us being wrong.

(c) The penalty. If $\hat{\Sigma}$ follows every random wiggle in S, it inherits S's error; if it ignores S, it inherits none.

Two sanity checks.

- Use $\hat{\Sigma} = S$ (no shrinkage): (a) $= 0$ and (c) $= 2V$, so the error is $0 - V + 2V = V$. Correct: the error of S is its own noise.
- Use a fixed $\hat{\Sigma} = F$ that ignores the data: (c) $= 0$, so the error is $\mathbb{E}\|F - S\|^2 - V = \|F - \Sigma\|^2$. Correct: a fixed guess is wrong by exactly its distance from the truth.

2. Plug in Ledoit–Wolf shrinkage

With $\hat{\Sigma} = \delta F + (1 - \delta)S$, the three pieces become

$$R(\delta) = \delta^2 D + (1 - 2\delta)V + 2\delta C,$$

where each letter is a single number:

Symbol	Definition	In words
D	$\mathbb{E}\|F - S\|_F^2$	how far the target is from the data
V	$\sum_{ij} \mathrm{Var}(s_{ij})$	how noisy the sample covariance is
C	$\sum_{ij} \mathrm{Cov}(f_{ij}, s_{ij})$	how much the target moves with that noise

$R(\delta)$ is a parabola in δ. Its lowest point is $\delta^* = (V - C)/D$: shrink more when S is noisy (V large), less when the target is far off (D large) or shares S's noise (C large). SURE replaces D, V, C by estimates from the data; that gives an estimated parabola we can minimize.

Why it is true (short proof; full walk-through on pages 5–6). Write the error as $(\hat{\Sigma} - S) + (S - \Sigma)$ and square. The squares give (a) and V; the cross term, using $\mathbb{E}s_{ij} = \sigma_{ij}$, is $2\sum \mathrm{Cov}(\hat{\sigma}_{ij}, s_{ij}) - 2V$. That is the three-piece formula. For LW, $\hat{\Sigma} - S = \delta(F - S)$ gives (a) $= \delta^2 D$, and $\mathrm{Cov}(\hat{\sigma}_{ij}, s_{ij}) = \delta\,\mathrm{Cov}(f_{ij}, s_{ij}) + (1-\delta)\,\mathrm{Var}(s_{ij})$ gives (c) $= 2\delta C + 2(1-\delta)V$. Adding: $\delta^2 D - V + 2\delta C + 2(1-\delta)V = \delta^2 D + (1-2\delta)V + 2\delta C$. □

3. Side by side with Stein

Same formula, different objects:

	Stein: estimate a mean	Thesis: estimate a covariance
Seen / wanted	X / θ	S / Σ
Right on average	$\mathbb{E}X = \theta$	$\mathbb{E}S = \Sigma$
(b) noise in the raw estimate	$p\sigma^2$	V
(a) how far we moved	$\lVert\hat{\theta} - X\rVert^2$	$\delta^2 \lVert F - S\rVert_F^2$
(c) copy-the-noise penalty	$2\sum_i \mathrm{Cov}(\hat{\theta}_i, X_i)$	$2\sum_{ij} \mathrm{Cov}(\hat{\sigma}_{ij}, s_{ij})$
How (c) is estimated	Stein's lemma (normal data only)	pairs of days (any data)
Shrink rule	$(1-\lambda)X$	$\delta F + (1-\delta)S$
Data-chosen amount	$\hat{\lambda} = p\sigma^2/\lVert X\rVert^2$	$\hat{\delta} = (\hat{V} - \hat{C})/\lVert F - S\rVert_F^2$

4. Where the thesis departs from Stein

Stein's way needs normal data. Stein's lemma turns the penalty (c) into a derivative, but only when the data follow a bell curve. Every earlier way of estimating V and C inherits that assumption: the Stein–Haff identity (Haff, 1980), Yi & Zou's Lemma 2, and RBLW (Chen et al., 2010). Stock returns have fat tails, so this matters.

The thesis way: count, don't assume. Both V and C are just combinations of fourth moments (here X_t is day t's return vector, mean known to be zero, $S = \frac{1}{T}\sum_t X_t X_t^\top$):

$$V = \frac{\mathbb{E}\lVert X\rVert^4 - \mathrm{tr}\,\Sigma^2}{T}, \qquad C = \frac{\mathrm{Var}(\lVert X\rVert^2)}{pT} = \frac{\mathbb{E}\lVert X\rVert^4 - (\mathrm{tr}\,\Sigma)^2}{pT} \quad (F = \nu I).$$

Each ingredient has a simple unbiased estimate:

Estimate	Formula	What it averages
$\hat{M} \to \mathbb{E}\lVert X\rVert^4$	$\frac{1}{T}\sum_t \lVert X_t\rVert^4$	each day's size, to the fourth power
$\hat{U}_2 \to \mathrm{tr}\,\Sigma^2$	$\frac{1}{T(T-1)}\sum_{t\neq u}(X_t^\top X_u)^2$	how alike two different days are
$\hat{U}_3 \to (\mathrm{tr}\,\Sigma)^2$	$\frac{1}{T(T-1)}\sum_{t\neq u}\lVert X_t\rVert^2\lVert X_u\rVert^2$	size of one day times size of another

Using pairs of *different* days ($t \neq u$) is the key: different days are independent, so the average of a product equals the product of averages, with no bias. Then

$$\hat{V} = \frac{\hat{M} - \hat{U}_2}{T}, \qquad \hat{C} = \frac{\hat{M} - \hat{U}_3}{pT}.$$

These are right on average for any distribution with finite fourth moments. So the SURE curve is exactly unbiased at every δ without assuming normality. The same works for the targets I (where $C = 0$) and D_S. It does *not* work for single-index or constant-correlation, because those targets are nonlinear in S.

Does it matter in practice? Yes. In the simulation ($p = 100$, AR(1), $r = 0.5$), SURE stays within about 3% of the best possible (oracle) error at every sample size, with normal or fat-tailed (t_{10}) data. With fat tails, RBLW and OAS, which assume normality, are 1.2–4.6 times the oracle error.

The role of Yi & Zou's Lemma

1. What the lemma says

Yi & Zou (2013), Lemma 1 (dissertation Lemma 2.2.1, eq. 2.6). For *any* estimator $\hat{\Sigma}$,

$$\mathbb{E}\|\hat{\Sigma} - \Sigma\|_F^2 = \mathbb{E}\|\hat{\Sigma} - S\|_F^2 - \sum_{i,j} \mathrm{Var}(s_{ij}) + 2\sum_{i,j} \mathrm{Cov}(\hat{\sigma}_{ij}, s_{ij}).$$

This is exactly the three-piece formula (a) − (b) + (c); pages 5–6 are its proof. Yi & Zou define $S = \frac{1}{n-1}\sum(X - \bar{X})(X - \bar{X})^\top$: the $n - 1$ divisor with an estimated mean, chosen precisely so that $\mathbb{E}S = \Sigma$ holds.

2. The role it plays

1. **It swaps an unknown for knowns.** The usual bias–variance split,

$$\mathbb{E}\|\hat{\Sigma} - \Sigma\|_F^2 = \sum_{i,j} \mathrm{Var}(\hat{\sigma}_{ij}) + \sum_{i,j} \big(\mathbb{E}\hat{\sigma}_{ij} - \sigma_{ij}\big)^2,$$

   contains the unknown σ_{ij} in the bias term. Yi's form replaces it with $\mathbb{E}\|\hat{\Sigma} - S\|_F^2$, which the data shows directly. That is what makes the risk estimable.

2. **It holds for any estimator.** Yi & Zou apply it to tapering; the thesis applies it to Ledoit–Wolf. Nothing needs re-deriving.

3. **It reduces the problem to two quantities**, $\mathrm{Var}(s_{ij})$ and $\mathrm{Cov}(\hat{\sigma}_{ij}, s_{ij})$. This is where the thesis and Yi & Zou part ways:

	Yi & Zou	Thesis
Estimator	tapering, $\hat{\sigma}_{ij} = w_{ij}s_{ij}$	LW, $\delta F + (1-\delta)S$
$\mathrm{Var}(s_{ij})$	Lemma 2.2.2: exact, normal data only	$(\hat{M} - \hat{U}_2)/T$, any distribution
Cov term	$w_{ij}\,\mathrm{Var}(s_{ij})$ (fixed weights)	needs $\mathrm{Cov}(f_{ij}, s_{ij})$ (F uses the data)

3. Rewriting the Frobenius error with the lemma

Step 1: apply the lemma to LW. With $\hat{\Sigma} = \delta F + (1-\delta)S$:

- (a): $\hat{\Sigma} - S = \delta(F - S)$, so $\mathbb{E}\|\hat{\Sigma} - S\|_F^2 = \delta^2 D$.
- (c): $\mathrm{Cov}(\hat{\sigma}_{ij}, s_{ij}) = \delta\,\mathrm{Cov}(f_{ij}, s_{ij}) + (1-\delta)\,\mathrm{Var}(s_{ij})$, so (c) $= 2\delta C + 2(1-\delta)V$.

$$\mathbb{E}\|\hat{\Sigma} - \Sigma\|_F^2 = \delta^2 D - V + 2\delta C + 2(1-\delta)V = \boxed{\delta^2 D + (1-2\delta)V + 2\delta C}.$$

Step 2: turn it into SURE. The risk $R(\delta) = \delta^2 D + (1-2\delta)V + 2\delta C$ has three unknown numbers, one per piece of the three-piece formula:

Piece	Unknown	In words
(a) how far we moved	$D = \mathbb{E}\|F - S\|_F^2$	distance between target and data
(b) noise in S	$V = \sum_{ij} \mathrm{Var}(s_{ij})$	how much S jumps around
(c) copying the noise	$C = \sum_{ij} \mathrm{Cov}(f_{ij}, s_{ij})$	how much the target jumps with S

SURE swaps each unknown for a number computed from the data that is right on average:

$$\mathrm{SURE}(\delta) = \delta^2 \hat{D} + (1-2\delta)\hat{V} + 2\delta \hat{C}.$$

Because δ is a fixed number we choose, $\mathbb{E}\,\mathrm{SURE}(\delta) = \delta^2 \mathbb{E}\hat{D} + (1-2\delta)\mathbb{E}\hat{V} + 2\delta\,\mathbb{E}\hat{C} = R(\delta)$. So the whole job is three estimates. Below, each one: what it measures, how to compute it, why it is right on average.

Setup

The data are T days of returns on p assets. Day t is a vector X_t (one return per asset), the days are independent, and the mean return is taken as 0. Then

$$S = \frac{1}{T}\sum_t X_t X_t^\top, \qquad \nu = \frac{\mathrm{tr}\, S}{p}, \qquad F = \nu I.$$

Two numbers per day or per pair of days are all we need:

$$n_t = \|X_t\|^2 \ \text{(size of day } t\text{)}, \qquad g_{tu} = X_t^\top X_u \ \text{(overlap of days } t \text{ and } u\text{)}.$$

The one idea behind (b) and (c)

A variance is "mean of the square minus square of the mean": $\mathrm{Var}(Y) = \mathbb{E}[Y^2] - (\mathbb{E}Y)^2$. From data:

- **Same-day products** $(Y_t \cdot Y_t)$ average to $\mathbb{E}[Y^2]$, the mean of the square.
- **Different-day products** $(Y_t \cdot Y_u,\ t \neq u)$ average to $(\mathbb{E}Y)^2$, the square of the mean, because independent days give $\mathbb{E}[Y_t Y_u] = \mathbb{E}Y_t \cdot \mathbb{E}Y_u$.

So: (average over same-day products) − (average over different-day products) is an unbiased variance. Pieces (b) and (c) are this idea applied to two different Y's.

Piece (a): $\hat{D}$

Measures: how far the target is from the sample covariance.
Compute: $\hat{D} = \|F - S\|_F^2 = \sum_{i,j}(f_{ij} - s_{ij})^2$.
Right on average: D is *defined* as the average of exactly this quantity, so nothing to prove.

Piece (b): $\hat{V}$

Measures: the noise in S. Each day contributes the matrix $Y_t = X_t X_t^\top$, and S is their average. So V is "how much one day's matrix varies", divided by T (averaging T independent days divides variance by T):

$$V = \frac{1}{T}\Big(\underbrace{\mathbb{E}\|Y\|_F^2}_{\text{mean of square}} - \underbrace{\|\mathbb{E}Y\|_F^2}_{\text{square of mean}} \Big) = \frac{1}{T}\Big(\mathbb{E}\|X\|^4 - \operatorname{tr}\Sigma^2\Big),$$

since $\|Y\|_F^2 = \|X\|^4$ and $\mathbb{E}Y = \Sigma$.
Compute: the matrix-version of "product" is $\langle Y_t, Y_u\rangle = g_{tu}^2$ (and $= n_t^2$ when $t = u$).

$$\hat{M} = \frac{1}{T}\sum_t n_t^2 \ \text{(same day)}, \qquad \hat{U}_2 = \frac{1}{T(T-1)}\sum_{t\neq u} g_{tu}^2 \ \text{(different days)}, \qquad \hat{V} = \frac{\hat{M} - \hat{U}_2}{T}.$$

Right on average: $\mathbb{E}\hat{M} = \mathbb{E}\|X\|^4$; for $t \neq u$, independence gives $\mathbb{E}g_{tu}^2 = \operatorname{tr}\Sigma^2$. So $\mathbb{E}\hat{V} = V$.

Piece (c): $\hat{C}$

Measures: how much the target moves together with S. $F = \nu I$ is zero off the diagonal (no movement there), and on the diagonal it is $\nu = \operatorname{tr} S/p$. Adding up the diagonal, $C = \operatorname{Cov}(\nu, \operatorname{tr} S) = \operatorname{Var}(\operatorname{tr} S)/p$. Now $\operatorname{tr} S$ is the average of the daily sizes n_t, so

$$C = \frac{1}{pT}\operatorname{Var}(n) = \frac{1}{pT}\Big(\underbrace{\mathbb{E}[n^2]}_{\text{mean of square}} - \underbrace{(\mathbb{E}n)^2}_{\text{square of mean}} \Big) = \frac{1}{pT}\Big(\mathbb{E}\|X\|^4 - (\operatorname{tr}\Sigma)^2\Big).$$

Compute: same idea with $Y = n$, a single number per day:

$$\hat{M} = \frac{1}{T}\sum_t n_t^2 \ \text{(same day)}, \qquad \hat{U}_3 = \frac{1}{T(T-1)}\sum_{t\neq u} n_t n_u \ \text{(different days)}, \qquad \hat{C} = \frac{\hat{M} - \hat{U}_3}{pT}.$$

Right on average: $\mathbb{E}\hat{M} = \mathbb{E}\|X\|^4$; for $t \neq u$, $\mathbb{E}[n_t n_u] = \mathbb{E}n_t\,\mathbb{E}n_u = (\operatorname{tr}\Sigma)^2$. So $\mathbb{E}\hat{C} = C$.

Worked example by hand ($p = 2$ assets, $T = 4$ days)

Days: $X_1 = (1, 1)$, $X_2 = (-1, 0)$, $X_3 = (0, 1)$, $X_4 = (-1, -1)$.

1. **Sizes** n_t: 2, 1, 1, 2.
2. **Overlaps** g_{tu} for the 6 pairs: $g_{12} = -1$, $g_{13} = 1$, $g_{14} = -2$, $g_{23} = 0$, $g_{24} = 1$, $g_{34} = -1$.
3. $S = \frac{1}{4}\sum_t X_t X_t^\top = \begin{pmatrix} 3/4 & 1/2 \\ 1/2 & 3/4 \end{pmatrix}$, so $\nu = \frac{3/4+3/4}{2} = \frac{3}{4}$ and $F - S = \begin{pmatrix} 0 & -1/2 \\ -1/2 & 0 \end{pmatrix}$.
4. **(a)** $\hat{D} = 0 + \frac{1}{4} + \frac{1}{4} + 0 = \frac{1}{2}$.
5. **Same day:** $\hat{M} = \frac{4+1+1+4}{4} = \frac{5}{2}$.
6. **(b) different days:** squares of overlaps $1 + 1 + 4 + 0 + 1 + 1 = 8$; each pair counts twice (t, u and u, t), so $\hat{U}_2 = \frac{2 \cdot 8}{4 \cdot 3} = \frac{4}{3}$, and $\hat{V} = \frac{5/2 - 4/3}{4} = \frac{7}{24}$.
7. **(c) different days:** products of sizes $2 + 2 + 4 + 1 + 2 + 2 = 13$, so $\hat{U}_3 = \frac{2 \cdot 13}{12} = \frac{13}{6}$, and $\hat{C} = \frac{5/2 - 13/6}{2 \cdot 4} = \frac{1}{24}$.
8. **SURE:** $\mathrm{SURE}(\delta) = \frac{1}{2}\delta^2 + (1 - 2\delta)\frac{7}{24} + 2\delta \cdot \frac{1}{24} = \frac{1}{2}\delta^2 - \frac{1}{2}\delta + \frac{7}{24}$.
9. **Best** δ**:** $\hat{\delta} = \frac{\hat{V} - \hat{C}}{\hat{D}} = \frac{7/24 - 1/24}{1/2} = \frac{1}{2}$, where SURE drops from $\frac{7}{24}$ (no shrinkage) to $\frac{1}{6}$.

On a computer

Form the $T \times T$ matrix $G = XX^\top$ (all overlaps; its diagonal is the sizes n_t). Then $\sum_{t \neq u} g_{tu}^2 = \sum_{t,u} G_{tu}^2 - \sum_t n_t^2$ and $\sum_{t \neq u} n_t n_u = (\sum_t n_t)^2 - \sum_t n_t^2$, so no loop over pairs is needed. In simulation ($p = 100$, $T = 20$, 4000 datasets) the average SURE matches the average actual loss at every δ (e.g. $\delta = 0.5$: 143.7 vs. 143.6).

Two cautions.

- SURE is right on average at each *fixed* δ. The chosen $\hat{\delta}$ is a ratio of random numbers, so it is not exactly right on average for the best δ^* (page 15). If $\hat{\delta}$ falls outside $[0, 1]$, clip it.
- Nothing above assumed normal data: only independent days, mean 0, and finite fourth moments.

Step 3: compare with Yi's own case. For tapering, $\hat{\sigma}_{ij} = w_{ij} s_{ij}$ with fixed weights, the same steps give

$$\mathrm{SURE} = \sum_{i,j} (1 - w_{ij})^2 s_{ij}^2 + \sum_{i,j} (2w_{ij} - 1)\widehat{\mathrm{Var}}(s_{ij}),$$

which is their eq. (2.11) up to an $(n-1)/n$ factor from how they scale S. LW toward $F = 0$ is the special case $w_{ij} = 1 - \delta$ for all (i, j), with $C = 0$.

4. A note on dropping V

The $-V$ term is the same for every estimator, so Yi & Zou drop it when only *comparing* estimators (their "partial risk"). For LW, V cannot be dropped entirely: it comes back through the Cov term via $(1-\delta)\,\mathrm{Var}(s_{ij})$. That is why V appears in $R(\delta)$ and in $\delta^* = (V-C)/D$.

A parallel worth checking

In both problems the data-driven intensity differs from a neighbouring estimator by a small finite-sample correction:

- Normal means: linear SURE gives $\hat{\lambda} = p\sigma^2/\|X\|^2$; James–Stein uses $(p-2)\sigma^2/\|X\|^2$. The -2 appears because James–Stein's SURE accounts for the intensity depending on X; plugging $\hat{\lambda}$ into a fixed-λ SURE does not.
- Covariance: LW (2004) uses $\frac{T-1}{T}\hat{V}/\|F-S\|_F^2$; SURE uses $(\hat{V}-\hat{C})/\|F-S\|_F^2$.

In both, SURE is unbiased for the risk at a fixed intensity, but the minimizer is a ratio of estimates and is not unbiased for the oracle. Open question: does a SURE that accounts for $\hat{\delta}$ depending on S, as James–Stein's does for $\hat{\lambda}$, give a further correction (the analogue of $p \to p-2$), and does it lower risk at small T?

Every piece of the thesis, mapped to Stein

Each result from the work with Professor Rosenman has a direct counterpart in Stein's normal-means problem. The table is the summary; the numbered items give the details (item 1 is derived on pages 5–6, item 6 is on page 15).

Thesis result	Stein counterpart	Status
1. Risk in SURE form (Yi & Zou Lemma 1)	Covariance-penalty form of SURE	Proved (pages 5–6, 10–11)
2. LW 2004 Pythagoras, $\delta^* = b^2/d^2$	James–Stein as empirical Bayes	Same identity
3. $R(\delta) < R(0)$ iff $0 < \delta < 2\delta^*$	JS dominates X iff $0 < c < 2(p-2)\sigma^2$	Proved, 2026-10-26
4. $\delta^* \geq 0$; clip $\hat{\delta}$ to $[0, 1]$	Positive-part James–Stein	Proved
5. Unbiased $\hat{V}, \hat{C}$ by U-statistics	Stein's lemma / Stein–Haff (Gaussian)	Proved for νI
6. LW = SURE minus finite-sample terms	p vs. $p-2$	Proved; $p-2$ analogue open
7. Gaussian check vs. Chen et al. eq. (7)	Gaussian SURE / Rao–Blackwell	Checked
8. Touloumis (2015) comparison	"Stein-type" shrinkage, U-statistics	To do
9. Single-index target	Stein's lemma on a nonlinear F	Open (target decision)
10. Simulations	`stein_js.py`	Done for νI

2. LW's Pythagoras is James–Stein's empirical-Bayes identity

Normal means: $\mathbb{E}\|X\|^2 = \|\theta\|^2 + p\sigma^2$, i.e. $d^2 = a^2 + b^2$ with $a^2 = \|\theta\|^2$ (true spread) and $b^2 = p\sigma^2$ (noise). The oracle linear intensity is

$$\lambda^* = \frac{p\sigma^2}{p\sigma^2 + \|\theta\|^2} = \frac{b^2}{d^2},$$

exactly LW (2004)'s $\delta^* = b^2/d^2$ on page 3. In the empirical-Bayes reading (Efron & Morris), with $\theta \sim N(0, \tau^2 I)$ the Bayes shrinkage factor is $\sigma^2/(\sigma^2 + \tau^2)$, and $(p-2)\sigma^2/\|X\|^2$ is an unbiased estimate of it. LW estimate b^2/d^2 consistently; the thesis estimates the numerator and denominator unbiasedly.

3. The risk-reduction window is the James–Stein window

- Linear normal means: $R(\lambda) - R(0) = \lambda\big[\lambda(p\sigma^2 + \|\theta\|^2) - 2p\sigma^2\big] < 0$ iff $0 < \lambda < 2\lambda^*$.
- James–Stein: $\mathrm{SURE}(c) - p\sigma^2 = \big(c^2 - 2c(p-2)\sigma^2\big)/\|X\|^2 < 0$ iff $0 < c < 2(p-2)\sigma^2$.

- Thesis: $R(\delta) - R(0) = \delta\big[\delta D - 2(V - C)\big] < 0$ iff $0 < \delta < 2\delta^*$, $\delta^* = (V - C)/D$.

All three are the same quadratic: any shrinkage up to twice the optimum beats no shrinkage. The window is set by the oracle; the estimated version $0 < \delta < 2\hat{\delta}$ only describes the estimated curve.

4. $\delta^* \geq 0$ and clipping are the positive-part story

$V \geq C$ by Cauchy–Schwarz on $\operatorname{tr} S$ (`sure_scaled_identity.pdf`), so $\delta^* \geq 0$: shrinking toward νI never makes the oracle want to push *away* from the target. But $\hat{\delta} = (\hat{V} - \hat{C})/\|F - S\|^2$ can fall outside $[0, 1]$, which is why LW clip it. That is the positive-part James–Stein estimator $\big(1 - (p-2)\sigma^2/\|X\|^2\big)_+ X$, which strictly improves on plain James–Stein (Baranchik, 1964). Clipping $\hat{\delta}$ should likewise only help, but it makes the SURE of the clipped rule no longer exactly the fixed-δ formula.

5. Unbiased terms: Stein's lemma vs. U-statistics

Stein estimates the covariance penalty by integration by parts, which needs the Gaussian density; its Wishart version (Stein–Haff) underlies Yi & Zou Lemma 2 and RBLW. The thesis gets the same three terms from moments: $\hat{V} = (\hat{M} - \hat{U}_2)/T$ and $\hat{C} = (\hat{M} - \hat{U}_3)/(pT)$ are unbiased under finite fourth moments alone. Monte Carlo agrees (`sims/out/results.txt`): e.g. t_{10}, $p = 100$, $T = 40$: $V = 340.24$ vs. mean $\hat{V} = 339.16$ (MC s.e. 0.86); $C = 0.944$ vs. 0.939 (s.e. 0.006).

7. The Gaussian check

Under Gaussian data the distribution-free decomposition gives an α^* identical to Chen et al. eq. (7), the oracle that RBLW estimates by Rao–Blackwellizing LW. So the thesis SURE contains the Gaussian Stein-world answer as a special case and drops the assumption it rests on. This is visible in the simulation: under Gaussian data SURE $\approx$ OAS; under t_{10} RBLW and OAS lose because they hard-code the Gaussian fourth moment.

8. Touloumis is "Stein-type" in the James–Stein sense

Touloumis (2015) titles the estimators "nonparametric Stein-type": James–Stein-style linear shrinkage toward νI, I, or D_S, with $\operatorname{tr}\Sigma^2$ and $\operatorname{tr}^2\Sigma$ estimated by unbiased U-statistics (Y_{1N}, Y_{2N}). That is the closest existing estimator to the thesis; the difference is that the thesis targets the finite-sample risk curve rather than a ratio-consistent intensity. The head-to-head comparison is still to do. Schäfer & Strimmer (2005), next on the reading list, is also James–Stein-type with term-by-term estimates.

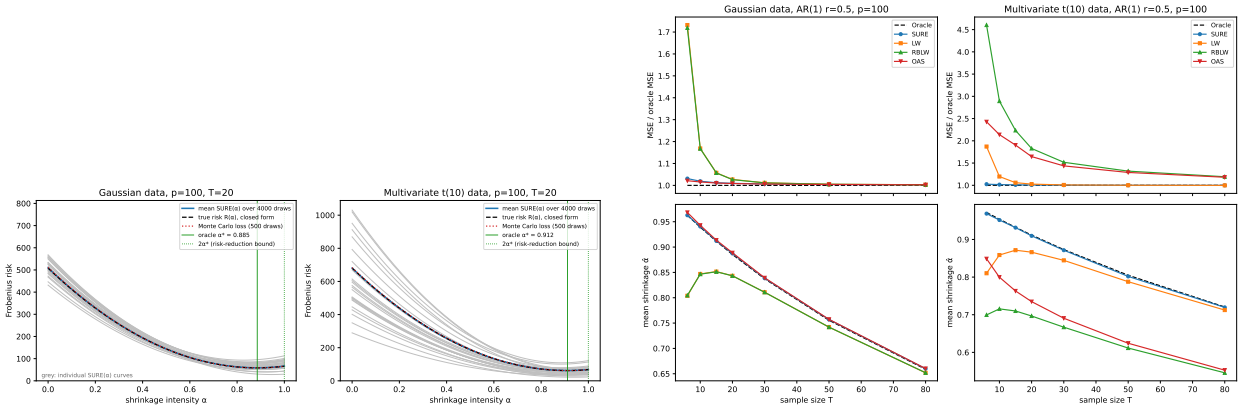
9. Single-index target: where Stein's lemma could come back

For F nonlinear in S (single-index, constant-correlation) no unbiased moment estimate of C is available. Under Gaussian data, Stein's lemma gives a route. With $x_t \sim N(0, \Sigma)$ and $\mathbb{E}[x_{ti} h(X)] = \sum_k \sigma_{ik}\, \mathbb{E}[\partial h / \partial x_{tk}]$, taking $h = x_{tj} f_{ij}$,

$$\mathrm{Cov}(f_{ij},\, x_{ti} x_{tj}) = \sum_k \sigma_{ik}\, \mathbb{E}\Big[x_{tj}\, \frac{\partial f_{ij}}{\partial x_{tk}}\Big],$$

and $\mathrm{Cov}(f_{ij}, s_{ij})$ is the average over t. This still involves Σ, so plugging in S gives an *approximate* SURE, not an exact one. This is a concrete option for the target decision to raise with Professor Rosenman: exact distribution-free SURE for linear targets, or Gaussian approximate SURE for single-index.

10. The simulations side by side



Left: SURE(δ) against the true $R(\delta)$ for νI ($p = 100$, $T = 20$): the mean curve is within 0.2% of the truth, the covariance version of the right panel on page 4. *Right:* MSE of SURE, LW, RBLW, OAS against the oracle. At $T = 6$: Gaussian, oracle 63.9, SURE 65.8, LW 110.6; t_{10}, oracle 70.5, SURE 72.2, LW 131.9, RBLW 324.3, OAS 171.1.

Where the work stands

1. **Question** (proposal, Aug 2026): can SURE tune δ, especially for targets with no closed-form δ?
2. **Literature**: Ledoit & Wolf (2003, 2004); Yi & Zou (2013); Chen, Wiesel, Eldar & Hero (2010); Touloumis (2015).
3. **Risk in SURE form**: $R(\delta) = \delta^2 D + (1 - 2\delta)V + 2\delta C$ (`sure_derivation.pdf`).
4. **Estimating the terms**: exact distribution-free $\hat{V}, \hat{C}$ for $F = \nu I$ (`sure_scaled_identity.pdf`).
5. **Theory**: $R(\delta) < R(0)$ iff $0 < \delta < 2\delta^*$, and $\delta^* \geq 0$. LW is SURE with two finite-sample terms dropped.
6. **Simulation**: `stein_js.py` (page 4) and `sure_sim.py` (SURE vs. LW, RBLW, OAS).
7. **Stein map**: every result above has a normal-means counterpart (pages 16–18).
8. **Open**: choice of target; S with $1/T$ vs. $1/(T-1)$; unknown mean; Touloumis comparison; portfolio application.

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